Assignment 2: Launching, Load balancing & Scaling on AWS EC2

Overview

This assignment provides hands-on experience with core AWS services and modern cloud computing patterns. You will start by launching and managing EC2 instances using AWS Console. Building on this foundation, you will configure load balancing and auto-scaling groups to understand how AWS handles fault tolerance and dynamic scaling under traffic. Finally, you will reflect on the trade-offs between AWS services to make informed architectural decisions for real-world cloud applications.

Deliverables- Each student must submit the following:

1. **Presentation Slides**
    - Title slide with student name(s), course, and assignment title.
    - Clear explanation of each completed task:
        - Task 1 (Access Module 6 Lab - 3 Introduction to Amazon EC2 from AWS academy account) – Launching an EC2 Instance using AWS console
        - Task 2 (Access **Module 6 Lab - 3 Introduction to Amazon EC2 from AWS academy account**) – Access AWS EC2 instance through console for monitoring, managing activities.
        - Task 3 (**Access Module 10 Lab - 6 Scale & Load Balance your Architecture from AWS academy account**) – Load Balancer & Auto Scaling Group setup and traffic generation
        - Task 4 – Reflection questions
    - Explain each step in PPT and include screenshots of key steps / results (e.g., launching setup, successful launch, running, terminating / Stop, monitoring and managing steps, load balancing, ASG scaling events).
    - Reflection question responses (must be included in slides).
    - Lessons learned and challenges encountered.
2. **Demonstration (in-class if possible)**
    - A 6–8-minute walkthrough of the slides and discussion of your results.

Presentation Guidelines

- Use professional formatting (legible fonts, consistent colors, organized layout).
- Ensure clarity: avoid overcrowding slides with text; use bullet points and visuals.

- Provide screenshots or videos as evidence of work completed.
- Keep presentations within the time limit (6–9 minutes).
- Be prepared to answer questions about the process and challenges.

1. Task Completion (70 points total)
Each AWS task is evaluated based on correctness, completeness, and evidence (e.g., screenshots output).

- Task 1 – Launch EC2 via Console
- Task 2 – Monitor and Manage activity such as status checks, troubleshoot, system log, CPU Utilization, Network In, Network Out, trigger the alarm, manage security group, resize the instance, stop and terminate the instance.
- Task 3 – Load Balancer & Auto Scaling Group. Such as Create an AMI for Auto Scaling, create a load balancer, create a launch template and an auto scaling group, verify that load balancing is working, test auto scaling, Terminate Web Server 1.
- Task 4 – Reflection Questions. Complete and thoughtful answers for all questions
    1. Why was termination protection enabled while launching the EC2 instance?
    2. Which metrics did you monitor to check the EC2 instance performance?
    3. Why did you modify the security group to allow HTTP access?
    4. What is the purpose of using a Load Balancer in this lab?
    5. What happens in the Auto Scaling group when application traffic increases?
    6. Why did you create an Amazon Machine Image (AMI) before creating the launch template?

2. Presentation Quality (20 points total)
- Clear structure (intro, results, discussion, conclusion)
- Slides visually organized and professional, for example, adding pictures.
- Screenshots as evidence

3. Professionalism & Delivery (10 points total)
- Confident and clear presentation delivery
- Ability to answer questions from peers/instructor

Submission Instructions
- Submit slides (PowerPoint or PDF) to Moodle by the deadline.

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